PRESIDENCY UNIVERSITY

Department of Computer Science and Engineering

CSE208 – DIGITAL LOGIC DESIGN LABORATORY

Laboratory Design Test

4-Bit Combined Adder–Subtractor

Carry, Borrow and Signed Overflow Detection

Mandatory IC:	**7483**, a 4-bit binary full-adder IC, physically used as the shared arithmetic core.
Design approach:	Open-ended: determine the additional logic, connections and implementation independently.
Required outputs:	Four-bit result, final carry, unsigned borrow and signed overflow.
Assessment:	**10 marks total**: Design (4), Simulation (2), IC minimization (2), Discussion questions (2).

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1. Design Challenge

Design, simulate, physically construct and evaluate a **four-bit combinational adder–subtractor** as an independently assessed design test (**10 marks**). The circuit shall perform either binary addition or binary subtraction, selected by a single input, without any manual rewiring. The **7483 IC is compulsory**. In addition to the arithmetic result, three independent status signals must be generated and verified: C_{out}, B_{out} and V.

This is a design test, not a circuit-following exercise. No ready-made gate-level architecture, interconnection method, status-flag Boolean equations, detailed pin map, or solved verification table is supplied. Students must research, design, justify and demonstrate their own implementation.

The arithmetic circuit must share the *same 7483 arithmetic core* for both operations. A separate adder and subtractor combined through an output multiplexer is **not acceptable**. The solution must be based on conventional fixed-function digital logic; a programmable microcontroller, FPGA, CPLD, ROM lookup table or prebuilt adder–subtractor module is not permitted as a replacement for the logic design.

2. Learning Outcomes

At the end of this assignment, students must be able to:

1. Independently design a combined binary arithmetic circuit around the 7483 IC.
2. Select and justify a common hardware approach for both addition and subtraction.
3. Interpret the same four-bit patterns as unsigned values and as signed two's-complement values.
4. Derive logic for the three status outputs, showing that they represent different arithmetic conditions.
5. Simulate and implement the design using available laboratory ICs.
6. Verify functionality with meaningful boundary cases and explain observed behavior.
7. Evaluate and improve hardware efficiency by reducing the number of IC packages used.

3. Black-Box Functional Specification

3.1 Inputs and outputs

Signal	Width	Required meaning
$A_3A_2A_1A_0$	4 bits	First binary operand, from MSB to LSB.
$B_3B_2B_1B_0$	4 bits	Second binary operand, from MSB to LSB.
M	1 bit	Operation selector (0: addition, 1: subtraction).
$R_3R_2R_1R_0$	4 bits	Least significant four bits of the selected arithmetic result.
C_{out}	1 bit	Actual final carry-out signal of the 7483 arithmetic operation.
B_{out}	1 bit	Unsigned-borrow indicator in subtraction; must be 0 in addition.
V	1 bit	Signed-overflow indicator for the selected operation.

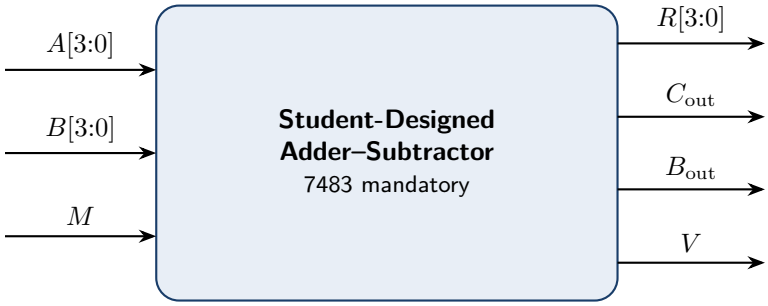


Figure 1: External interface only. Internal components, signal processing and interconnections are deliberately unspecified.

3.2 Arithmetic behavior

Mode	Operation	Required result
$M = 0$	Addition	$R = (A + B) \bmod 16$
$M = 1$	Subtraction	$R = (A - B) \bmod 16$

For example, R always consists of exactly four bits, regardless of whether the mathematical answer needs more bits. Students must independently decide how to use the 7483 to realize both modes.

4. Precise Status-Signal Requirements

The three flags have **distinct meanings**. They must be displayed and verified independently.

4.1 Final carry-out C_{out}

C_{out} must represent the final carry produced by the **actual 7483 operation**. It must not be replaced by the borrow or overflow signal. During addition, it indicates a carry beyond the 4-bit unsigned word. In the chosen standard two's-complement subtraction convention, it indicates whether an unsigned borrow *did not* occur. Students must explain and derive this distinction in their reports.

4.2 Borrow-out B_{out}

B_{out} is asserted precisely when the unsigned subtraction $A - B$ requires a borrow beyond the most significant bit. During addition ($M = 0$), B_{out} must remain zero for all inputs. Students must derive a correct circuit satisfying these conditions.

4.3 Signed overflow V

V must be asserted precisely when the *correct signed* arithmetic answer is outside the range representable by four-bit two's-complement numbers:

$$\boxed{-8 \leq \text{signed integer} \leq +7.}$$

Signed overflow is *not equivalent* to a carry or a borrow. Students must find, derive and justify a suitable digital logic implementation of V that works in both operation modes.

Important interpretation rule. The same operand bit pattern may be viewed as unsigned or signed. B_{out} and the carry interpretation concern *unsigned* arithmetic; V concerns *signed* arithmetic. Neither may substitute for the other. The mathematical meaning of each flag must be tested, not merely asserted in the report.

5. Mandatory Hardware and Design Constraints

1. **7483 IC:** Exactly one 7483-family unit supplied/approved by the laboratory must act as the common 4-bit arithmetic core. **Using a 74283 or 74HC283 in its place is not permitted**.
2. **Single design:** Both operations must use the same assembled circuit and the same 7483. Switching M must not involve moving any wire or replacing any IC.
3. **Independent design:** Students shall determine the additional logic types and quantities, circuit interconnections, Boolean expressions and pin assignments. No specific supporting-gate combination is prescribed.
4. **IC-based gates are compulsory:** Every required logic gate or function shall be implemented through an appropriate *physical, standard fixed-function logic IC* (normally from the approved 74xx/74LS families), in addition to the mandatory 7483. Drawn logic symbols by themselves are not a hardware implementation. Standalone virtual gates in a simulator are insufficient unless each is explicitly assigned to a real IC package and its pin connections in the final schematic. Do not realize logic gates using discrete transistors, diodes, or manual switch tricks. Programmable controllers and programmable logic devices are excluded.
5. **Gate-package mapping:** For every gate/function in the final circuit, identify the exact relevant IC part number, the gates used within the package, and their connection/pin assignments. Multiple gates may share one IC package where electrical requirements permit; this is encouraged for minimization.
6. **Required controls:** Four independently settable bits for A, four for B, and one for M.
7. **Observable outputs:** Four result bits and the three flags must all be independently observable, for example through suitably driven LEDs or logic indicators.
8. **Electrical correctness:** Use regulated power compatible with the exact devices; defined logic levels, sound grounding, safe LED loading and decoupling are mandatory. No input may be left undefined.
9. **Datasheet verification:** Students must identify the manufacturer's exact IC code and use its corresponding pin diagram. *The 7483/74LS83A pin arrangement is not the same as the 74283 arrangement*; do not assume one data sheet applies to the other.
10. **Hidden carry nodes:** Students must not assume the 7483 exposes all intermediate carry signals as pins. Their chosen overflow-detection design must use signals actually accessible in the constructed circuit.

6. Design Deliverables Before Construction

Each student or assigned group must prepare:

1. **Design proposal:** A short explanation of the proposed arithmetic approach and its justification.
2. **Logic derivations:** Derive Boolean expressions for any additional input-conditioning/control logic and all three status outputs. Use truth tables or Karnaugh maps wherever useful.
3. **Block diagram:** Show the student's own data flow and the placement of the mandatory 7483.
4. **Gate-level schematic:** Include all gates, signal labels, logic connections, input switches and output indicators.
5. **Wiring plan:** State each IC part number, relevant pin connections and supply connections from verified datasheets.
6. **IC budget and gate mapping:** State the planned total number of physical IC *packages*, identify each supporting gate's package, and explain opportunities for minimization.

7. Implementation and Validation

7.1 Design and simulation stage

1. Obtain and study the correct 7483 data sheet.
2. Devise an implementation meeting the full black-box specification. Do not use a ready-made combined adder–subtractor block.
3. Simulate the complete design, including observable status flags. Use device/package models if available; otherwise show explicit mapping from every simulated gate to its real fixed-function IC package and pin connections.
4. Verify both modes and all compulsory test vectors in Section 8.
5. Refine the circuit when the result or any flag is incorrect. Record the revision and reason.

7.2 Breadboard or trainer-kit stage

1. Assemble the approved student-designed circuit, using the mandatory physical 7483 IC.
2. Check correct pin orientation, voltage levels, supply bypassing, LED current limits and signal definitions *before* switching on power.
3. Test addition and subtraction without changing any hardware connections.
4. Verify R, C_{out}, B_{out} and V for the compulsory vectors and two original vectors.
5. Demonstrate the working circuit and explain the reasons for the chosen design and IC count.

8. Compulsory Test Vectors

For *every* row, calculate the expected output bits and all three flag values independently; compare them with simulated and hardware observations. No answers are provided here.

Test	M	A	B	Test	M	A	B
T1	0	0010	0011	T8	1	0011	0101
T2	0	0101	0010	T9	1	0111	1101
T3	0	0111	0011	T10	1	1000	0001
T4	0	1001	1101	T11	1	0000	0000
T5	0	1111	0001	T12	1	0000	0001
T6	0	0000	0000	T13	1	1000	1001
T7	1	0101	0011	T14	1	1010	0101

Two original tests: Specify one additional vector for addition and one for subtraction, distinct from those above. Give a technical justification for the pair selected.

8.1 Observation table

Use one row per test and *separate* the calculated, simulated and hardware measurements. A compact suggested format follows; repeat as needed.

Test	M	A	B	Source	R	C_{out}	B_{out}	V / remarks
T___				Theory				
				Simulation				
				Hardware				
T___				Theory				
				Simulation				
				Hardware				

Table 1: Repeat the three-row record for all 16 required tests (14 supplied and 2 original).

A test is successful only if the four result bits and *all three* flag outputs match the independently determined theoretical values. Identify any mismatch and explain its correction.

9. IC Minimization Challenge (2 Marks)

Optimization objective: Meet the entire black-box specification using as few *physical IC packages* as possible. The 7483 is compulsory and counts as one package. Gate-level functions must be built using relevant ICs; eliminating hardware by replacing actual gates with simulation-only primitives, programmable devices, or an unapproved circuit is not a valid minimization.

9.1 Counting rules

- Count **each physical logic IC package used** in the final operating circuit, including the required 7483 and every supporting gate/logic IC. Count a multi-gate package *once*, not once per gate.
- Do not count wires, breadboard, switches, LEDs, resistors, capacitors, external power supply or passive input/output accessories as IC packages.
- Supply an **IC inventory** listing every exact part number, quantity, gate utilization and total package count, with a legible final circuit photo and schematic matching that inventory.
- Only circuits that satisfy *all required operations and flags*, use the mandatory 7483 and physical IC-based gates, and pass the compulsory validation tests are eligible for minimization marks. A simulation-only design is ineligible for these two marks.

9.2 Mark allocation (maximum 2 marks)

Let N_{min} be the smallest **verified** number of IC packages among eligible, correctly working submissions in the assessed section, and let N be the count in an individual student's verified design.

Verified physical IC package count	Marks
$N = N_{min}$	**2**
$N = N_{min} + 1$	**1**
$N \geq N_{min} + 2$	**0**

Tied minimum-count designs receive the same marks. Students must show that their count is legitimate and fully functional; unsuccessful or unverified designs receive **0/2** for minimization. This is *part of* the 10-mark assessment, **not a bonus** added to the total. Unused gates do not reduce

the package count, so students should consider using existing gates within a package before adding another IC.

10. Laboratory Report Format

The report must include the following sections in order:

1. **Cover page:** university, course, experiment title, student name/ID, section, instructor and date.
2. **Objectives and scope:** describe the design problem and selected operation modes.
3. **Theory:** fixed-width arithmetic, signed/unsigned interpretation, the 7483 function and three status conditions.
4. **Logic derivations:** derive and simplify any logic required for the two modes and all three flags. Clearly define every signal used.
5. **Hardware details:** exact IC part numbers, mapping of every used gate to a real logic IC and its pins, connection table, electrical checks and package count.
6. **Original circuit design:** explain the selected approach; include an original block diagram and complete gate-level circuit schematic.
7. **Simulation:** simulator name, screenshots and explanations of the tested modes and flag outputs.
8. **Results and verification:** 14 given and 2 student-selected vectors, theoretical/simulated/observed comparisons and discrepancy analysis.
9. **Discussion questions (2 marks):** answer both questions in Section 12 using evidence from your design and test results.

11. Assessment Rubric (Total: 10 Marks)

Assessment component	Marks
Design: Original shared-7483 architecture (1), correct result/three-flag logic (1), real relevant ICs and sound hardware construction (1), complete annotated schematic/derivation and IC mapping (1)	**4**
Simulation: Functional verification of both modes and required test vectors (1); readable simulations/screenshots documenting result bits and all flags, with comparison to theory (1)	**2**
IC minimization: Scored using verified physical IC-package count and eligibility rules in Section 9	**2**
Discussion question 1: Evidence-based distinction between carry, borrow and signed overflow	**1**
Discussion question 2: Justification and verification of the student's overflow-detection approach with the available 7483 connections	**1**
Total	**10**

Marking note: The observation tables, design report and hardware demonstration support the criteria above; they do not create extra marks. The specified 7483 and suitable physical gate ICs are mandatory. A missing or incorrectly implemented status flag means the required design is incomplete and prevents eligibility for the minimization marks.

12. Two Compulsory Discussion Questions (2 Marks)

Answer both questions in your lab report. Each question carries 1 mark. Support your answers with evidence from your own design and verification results.

Question 1. Carry, borrow and overflow (1 mark): Why are C_{out}, B_{out} and V not interchangeable in a 4-bit adder–subtractor? Use two of your verified subtraction test cases: one showing unsigned borrow without signed overflow, and another showing signed overflow without unsigned borrow. How do you interpret each case as signed and unsigned arithmetic?

Question 2. Signed-overflow detection using 7483 (1 mark): How does your own 7483-based circuit detect signed overflow in *both addition and subtraction modes*, considering that not all internal carry signals are available as 7483 output pins? How did you verify that your chosen logic correctly distinguishes overflow from non-overflow?

Non-negotiable acceptance conditions. Use the physically installed 7483 as the shared arithmetic core; support both operations without rewiring; produce and correctly demonstrate R, C_{out}, B_{out} and V. If any status signal is missing or incorrectly implemented, the required circuit is incomplete and the design is ineligible for minimization marks.